

Mapping & MIDI

Programming and Music
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Assignment 3

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The third assignment of the year contains two parts:

1. A **generative system** that can create sound material and is based on a graphical user interface and/or a midi control setup.
2. A **short etude** that is composed using the system.

The code should be handed in as well as a small text with a minimum of **300 words** explaining the approaches and the value these could have in a musical setting.

The assignment should be handed in before the the **5th of June**. (and will be discussed during this class). In the classes of **8th** of May you should already present me the direction and progress of the project. The best would be to start soon.

Assignment 3, System & GUI

The generative system must be capable of the following requirements:

- Generate *musical material* that can be used in a musical composition or live performance.
- Contain a **GUI** or use **MIDI** for its main controls.
- Create sound structures based on *reading samples* and/or *sound synthesis*.
- **Process** the generated audio with built in *effects* or sound *transformation processes*.

Assignment 3, Etude

The small composition must meet the following requirements:

- Be at least **3 minute** long.
- Use **material** created by the generative system.
- Exist as an attempt to create **something of value** to its composer (not be simply a demo or rough sketch).
- Putting the material together in another program is allowed but could also be done in SuperCollider.

Mapping

Mapping

The process of transferring control values to musical parameters. Is often discussed in terms of four different categories:

- * *One to One*

- * *One to Many*

- * *Many to One*

- * *Many to Many*

Mapping

There exists endless possibilities of connecting information from external processes or controls to musical parameters. It is very important to consider the **mapping** carefully and in many cases the mapping itself is where the magic happens.

In the field of algorithmic composition mapping is less discussed than properties or behavior of generative algorithms.

It is sometimes assumed mapping is personal or full of irrelevant details.

Mapping

Mapping is still what connects external controls to musical parameters and can be considered to be part of ***compositional choices*** a composer has to make.

When mapping choices are made part of a system the system can be thought of as a ***compositional system*** or ***composed instrument***.

Mapping

Compositional choices are focused on *planning* but also on setting *limits* and creating *boundaries*. This overlaps somehow with instrument design and it is an interesting to discover where one considers his mapping to exist.

Connecting sound processes to compositional choices creates an *abstraction* that is essential to mapping procedures. What is mapped and how the behavior of the mapping is implemented has an important effect on choices made both for sound generation and how ones compose for a given system.

Considerations

- * Does a mapping make sense intuitively ?
- * Does the mapping curve respond well to its domain ?
- * Is the control range expressive enough ?
- * Does the control range contain the right space for change ?
- * Can the mapping effect be easily remembered ?
- * Does a mapping relate well to other mappings in the system ?
- * How isolated or dependent is a mapping to other controls ?

Scaling and Offsetting

Scaling a quantity means to *change its size*, or magnitude, by multiplication. Scaling an object visually would mean to make it appear bigger or smaller without altering its origin.

Offsetting a quantity means to add a value to it. Offsetting affects the origin of something without affecting the size.

Scale and Offset are important mapping procedures and many UGens in SuperCollider take these as standard arguments.

Curves and Envelopes

Transferring one range of values to another can either be done *linearly* or by using a *transfer function*.

The envelope is a very simple yet powerful data structure in sound synthesis and can be used also for parameter mapping. Using envelopes one can define *custom transfer functions* from one domain to the other.

A *normalized envelope* is an envelope where its values have a sum of 1. Using a normalized envelope is recommended since the range can easily be changed independently of the envelope values.

Processing

In addition to mapping curves or procedures, data can also be both ***pre-processed*** and/or ***post-processed***.

Pre-processing tasks could look at incoming data and ***remove*** or ***replace*** certain items. For big data sets it could also ***select*** or ***reject*** items based on a given criteria.

Post-processing tasks would involve treatment of data such as making values ***last*** or ***lag*** for some duration. ***Delay*** on control signals could also be introduced as well as ***interpolation*** between new and old values.

(from ***Mapping and visualization with SuperCollider***

<http://marinoskoutsoomichalis.com/mapping-and-visualization/>)

MIDI

Information about MIDI support in SuperCollider can be found in the help file *UsingMIDI* and the help file *MIDI*.

Using *MIDIFunc*, functions can be defined to respond to received MIDI messages. Several convenience methods are available to deal with specific MIDI messages such as *noteOn*, *cc* etc.

The *MIDIClient* class handles MIDI connection and communication with the operating system. It has to be initialized and once functions have been registered they have to be freed if they should no longer be called.

Exercises

Exercises - MIDI & Mapping

- 1. Create a one to many mapping for a synthesis process where the transfer function is different from the standard linear one.*
- 2. Create a mapping where the results are not audible until at least several seconds after the input control has been stopped.*
- 3. Create a mapping where an incoming signal is used to control a pattern and a synth at the same time.*
- 4. Create a mapping using two trajectories for one input signal where the result is a random number between the two (a tendency mask).*

